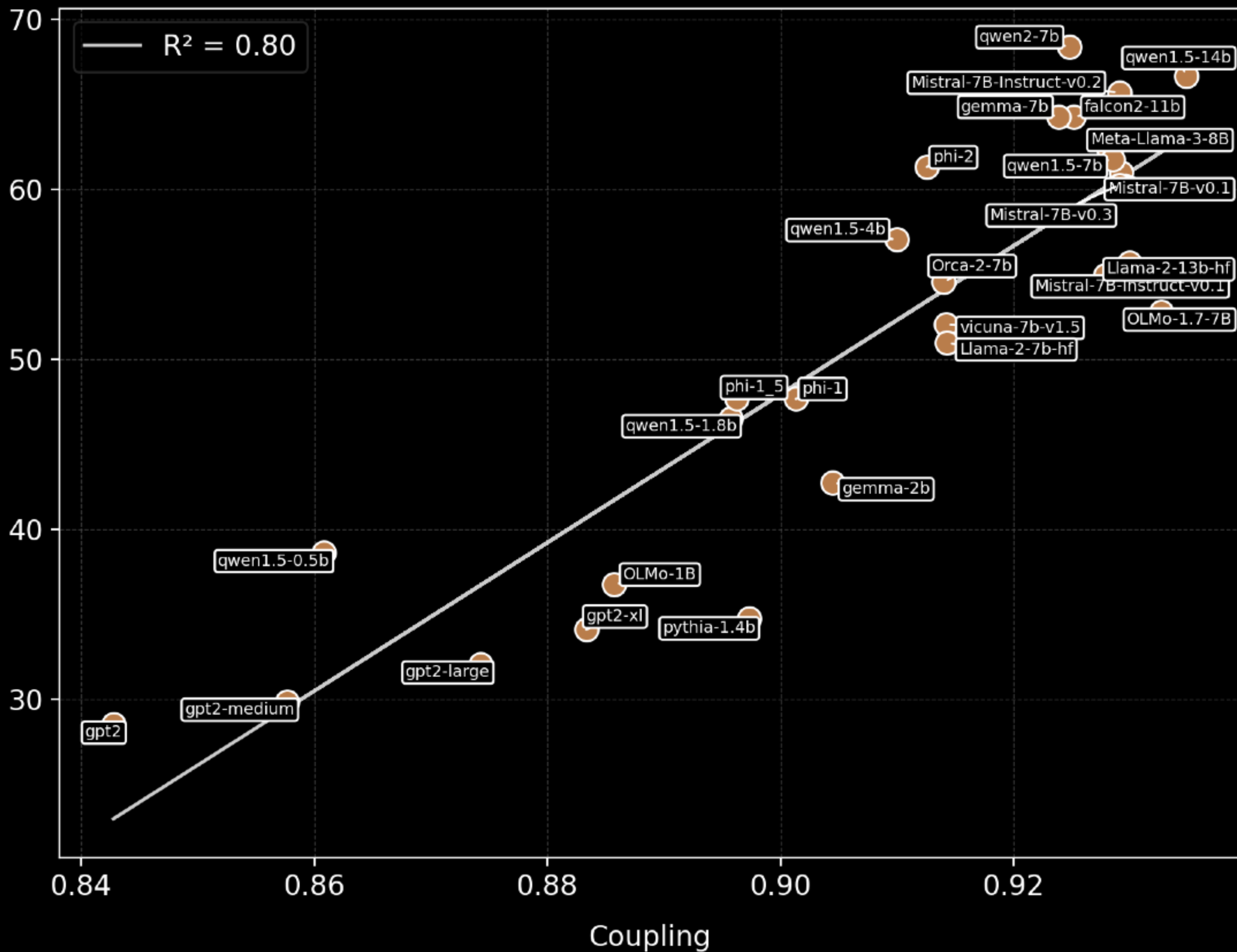


LLM Benchmark Score

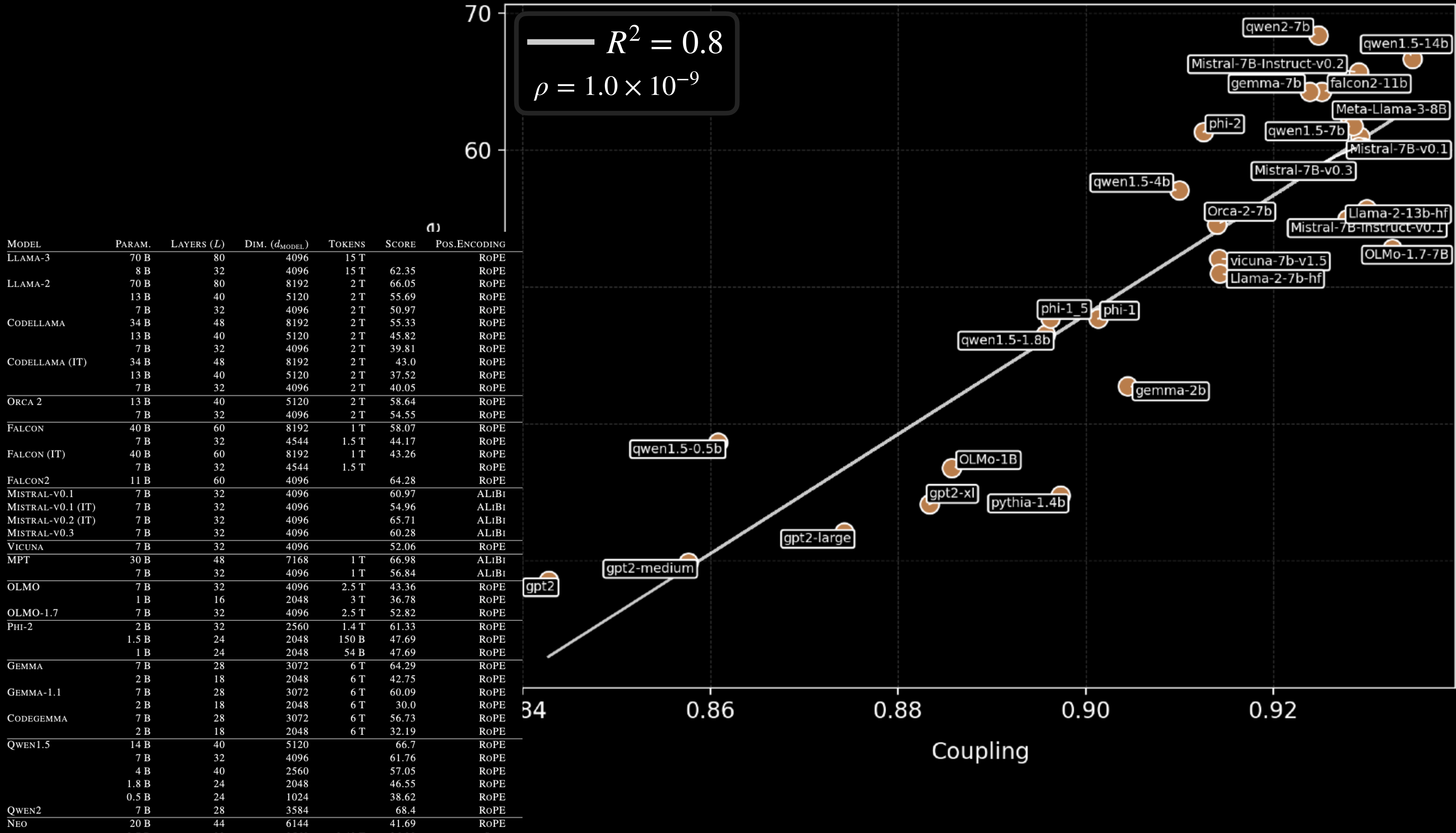



$$R^2 = 0.8$$

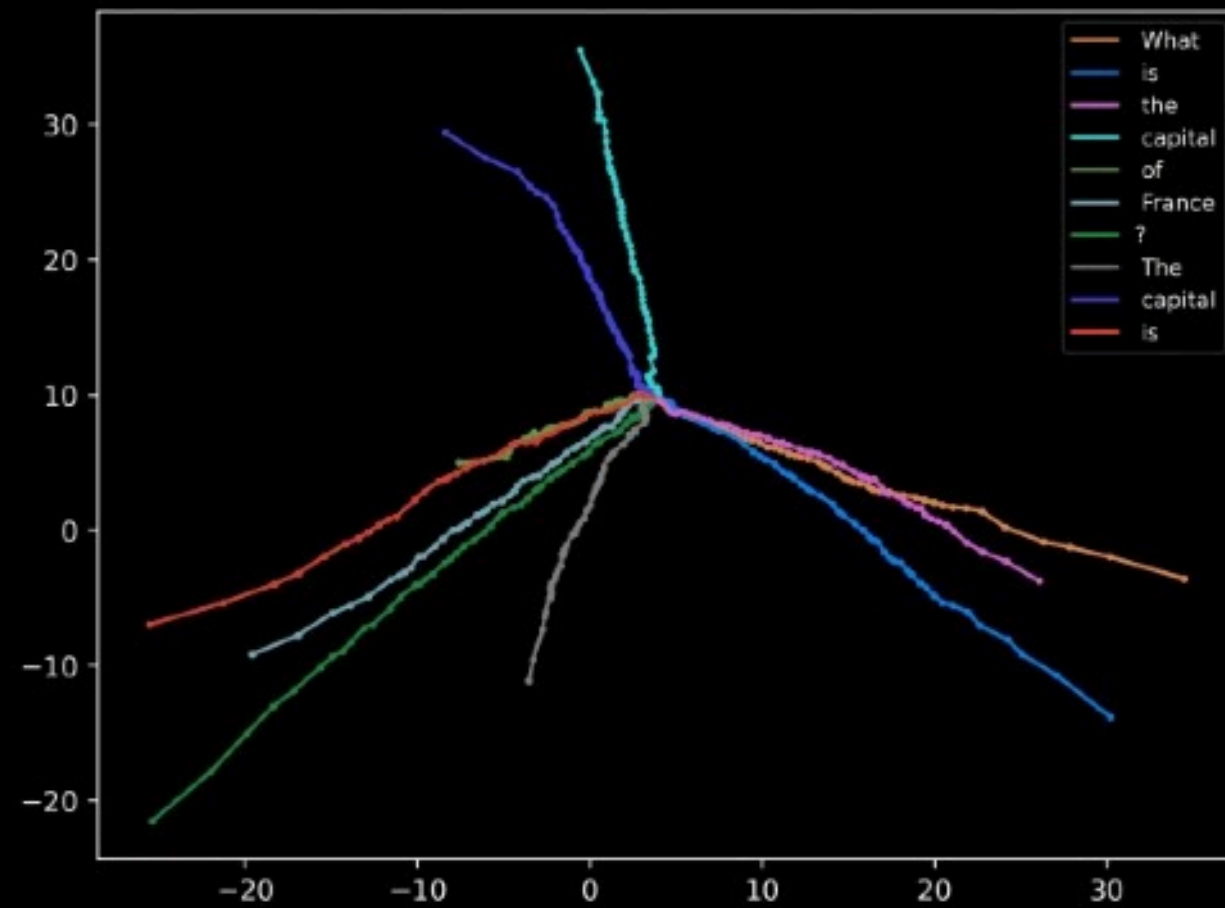
$$\rho = 1.0 \times 10^{-9}$$

MODEL	PARAM.	LAYERS (L)	DIM (d_{MODEL})	TOKENS	SCORE	POS.ENCODING
LLAMA-3	70 B	80	4096	15 T		RoPE
	8 B	32	4096	15 T	62.35	RoPE
LLAMA-2	70 B	80	8192	2 T	66.05	RoPE
	13 B	40	5120	2 T	55.69	RoPE
	7 B	32	4096	2 T	50.97	RoPE
CODELLAMA	34 B	48	8192	2 T	55.33	RoPE
	13 B	40	5120	2 T	45.82	RoPE
	7 B	32	4096	2 T	39.81	RoPE
CODELLAMA (IT)	34 B	48	8192	2 T	43.0	RoPE
	13 B	40	5120	2 T	37.52	RoPE
	7 B	32	4096	2 T	40.05	RoPE
ORCA 2	13 B	40	5120	2 T	58.64	RoPE
	7 B	32	4096	2 T	54.55	RoPE
FALCON	40 B	60	8192	1 T	58.07	RoPE
	7 B	32	4544	1.5 T	44.17	RoPE
FALCON (IT)	40 B	60	8192	1 T	43.26	RoPE
	7 B	32	4544	1.5 T		RoPE
FALCON2	11 B	60	4096		64.28	RoPE
MISTRAL-v0.1	7 B	32	4096		60.97	ALiBi
MISTRAL-v0.1 (IT)	7 B	32	4096		54.96	ALiBi
MISTRAL-v0.2 (IT)	7 B	32	4096		65.71	ALiBi
MISTRAL-v0.3	7 B	32	4096		60.28	ALiBi
VICUNA	7 B	32	4096		52.06	RoPE
MPT	30 B	48	7168	1 T	66.98	ALiBi
	7 B	32	4096	1 T	56.84	ALiBi
OLMO	7 B	32	4096	2.5 T	43.36	RoPE
	1 B	16	2048	3 T	36.78	RoPE
OLMO-1.7	7 B	32	4096	2.5 T	52.82	RoPE
PHI-2	2 B	32	2560	1.4 T	61.33	RoPE
	1.5 B	24	2048	150 B	47.69	RoPE
	1 B	24	2048	54 B	47.69	RoPE
GEMMA	7 B	28	3072	6 T	64.29	RoPE
	2 B	18	2048	6 T	42.75	RoPE
GEMMA-1.1	7 B	28	3072	6 T	60.09	RoPE
	2 B	18	2048	6 T	30.0	RoPE
CODEGEMMA	7 B	28	3072	6 T	56.73	RoPE
	2 B	18	2048	6 T	32.19	RoPE
QWEN1.5	14 B	40	5120		66.7	RoPE
	7 B	32	4096		61.76	RoPE
	4 B	40	2560		57.05	RoPE
	1.8 B	24	2048		46.55	RoPE
	0.5 B	24	1024		38.62	RoPE
QWEN2	7 B	28	3584		68.4	RoPE
NEO	20 B	44	6144		41.69	RoPE
	2.7 B	32	2560	0.42 T	36.20	SINE
PYTHIA	12 B	36	5120	0.3 T	38.82	RoPE
	6.9 B	32	4096	0.3 T	39.30	RoPE
	2.8 B	32	2560	0.3 T	37.09	RoPE
	1.4 B	24	2048	0.3 T	34.75	RoPE
	1 B	16	2048	0.3 T	32.78	RoPE
GPT-2	1.5 B	48	1600		34.12	SINE
	774 M	36	1280		32.07	SINE
	355 M	24	1024		29.87	SINE
	117 M	12	768		28.53	SINE

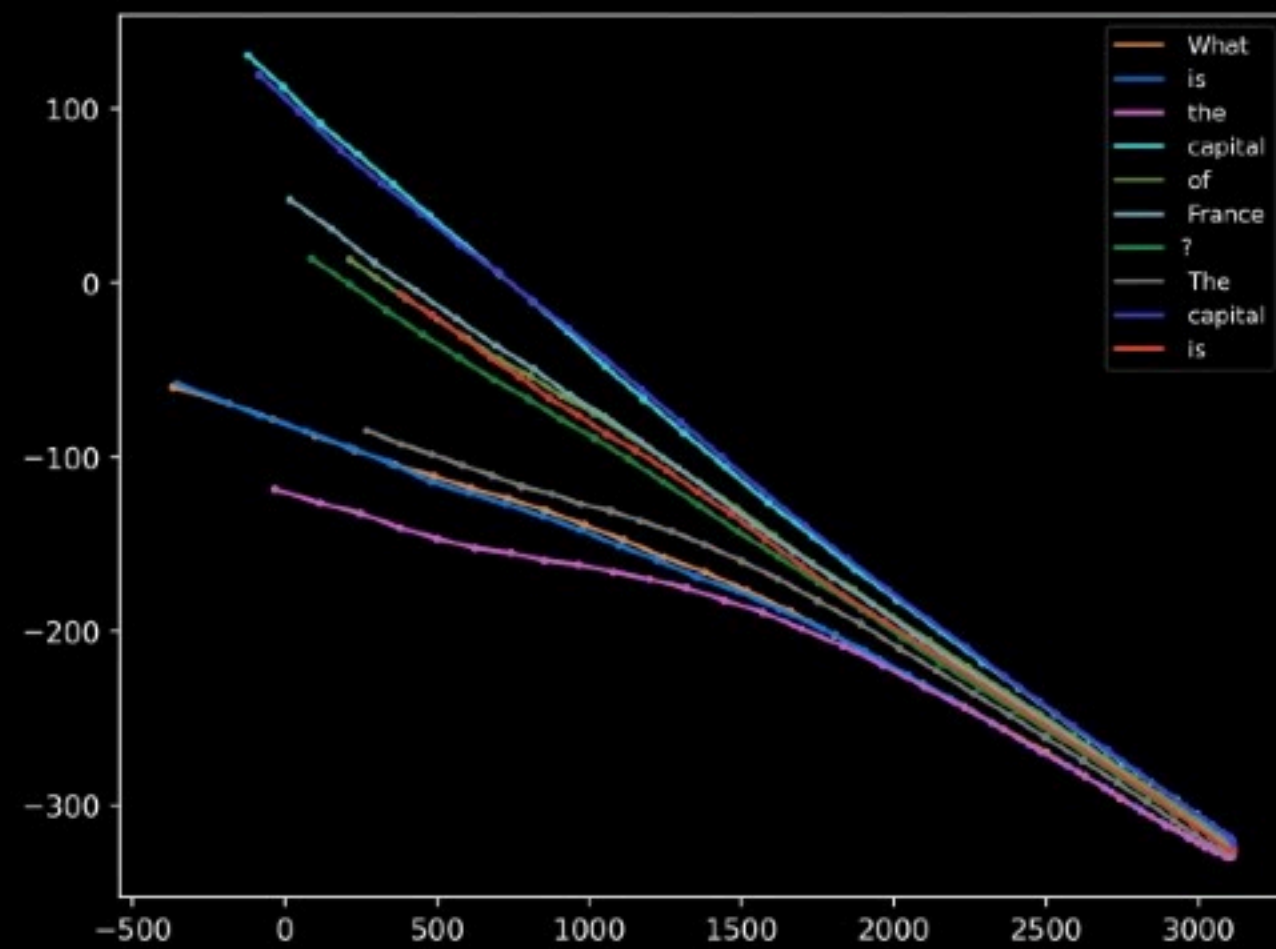
Robustness & Correlation with Generalization



Robustness & Correlation with Generalization



(a) Llama-3 70B



(b) MPT 30B

